Andrii Iarmolenko

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Graduating Mechanical Engineering student (May 2027) focused on robotics, hardware design, and electromechanical sensor integration. Experienced in CAD modeling, wearable exoskeleton research, and aerospace prototyping, backed by hands-on professional trade experience as a plumber mechanic.

EDUCATION

The City College of New York (CCNY) — New York, NY **Expected May 2027**
Bachelor of Engineering in Mechanical Engineering | **GPA: 3.7** / **4.0**

- **Relevant Coursework:** Finite Element Analysis (FEA), Fluid Mechanics, Heat Transfer, Manufacturing Processes, Computer-Aided Design (CAD), Materials Science

Borough of Manhattan Community College (BMCC) — New York, NY **Sept 2022 – Aug 2024**
Associate of Engineering Science | **GPA: 4.0** / **4.0**

TECHNICAL SKILLS

- **Software & Tools:** SolidWorks, Fusion 360, AutoCAD, MATLAB, LabVIEW, Zemax, Python, C++, CAM
- **Hardware & Instrumentation:** ESP32, Arduino, Load Cells, Force-Sensitive Resistor (FSR) Arrays, Analog Multiplexers (CD74HC4067), Oscilloscopes, Multimeters, RC Telemetry, Soldering, Wire Harnessing
- **Engineering & Fabrication:** 3D Printing, CAD Modeling, Finite Element Analysis (FEA), Experimental Design, Data Acquisition (DAQ) & Signal Processing

EXPERIENCE

Mechanical Test Engineer — AIAA Design-Build-Fly, CCNY *Sept 2026 – Present*

- Built an instrumented **motor dynamometer test bench** using modular aluminum extrusions and a **load cell**; programmed an **ESP32** to log thrust, RPM, and voltage across 0–100% throttle sweeps.
- Selected optimal motor and propeller configurations by analyzing **thrust-to-weight test data**, increasing static thrust by **18%** while mitigating thermal issues.
- Designed and wired a **central power distribution board** and **RC telemetry interface** to remotely control payload electronics during flight testing.

Research Assistant — BIOWEAR Lab, CCNY *Oct 2025 – July 2026*

- Designed a **16-channel multiplexed Force-Sensitive Resistor (FSR) array** on an **ESP32** to measure plantar pressure distributions for an **assistive ankle exoskeleton**.
- Integrated **CD74HC4067 analog multiplexers** to route **16 sensors** to a single ADC channel, cutting wiring complexity and mechanical strain at the robotic joint.
- Implemented **hardware pull-down networks** and **digital filtering** to eliminate floating-ground noise, achieving stable gait force capture at **100 Hz**.
- Modeled and **3D-printed** impact-resistant **sensor enclosures** in **SolidWorks** to withstand repeated high-load foot strikes.
- Built a **Python serial data collection pipeline** with automated timestamping, resolving COM port handshake issues for **lossless logging** during extended testing.

Research Student — MobilizeGreen Fellowship, CCNY *June 2025 – Aug 2025*

- Simulated and optimized **LiDAR optical systems** in **Zemax** and **MATLAB** to reduce laser beam divergence and improve target resolution.
- Validated simulated optical models against physical **benchtop ray-path measurements**.

Research Student — NSF REU Program, CCNY *June 2024 – May 2025*

- Improved **LiDAR accuracy** by integrating **Varioptic liquid lenses** into an adaptive **dual-lens optical system**.
- Automated multi-axis optical test sequences in **LabVIEW**, cutting experimental **data collection cycle time**.
- Refined **MATLAB signal-processing scripts** for reliably detecting small, fast-moving targets.
- Produced **technical reports** and delivered weekly research presentations for a multidisciplinary engineering team.

Plumber Mechanic — Real Plumbing Corp., New York, NY *Feb 2015 – June 2018*

- Earned promotion from apprentice to mechanic within **12 months** based on technical diagnostic capability and field execution.
- Diagnosed and resolved complex **flow, pressure, and thermal faults** in high-pressure commercial and residential piping systems.
- Led **system commissioning** and **hydrostatic pressure testing** of commercial piping assemblies.
- Coordinated daily on-site installations with **cross-functional teams** (general contractors, electricians, and HVAC) to keep multi-week schedules on track.

CONFERENCE PRESENTATIONS

- **Emerging Researchers National Conference (ERN 2025):** *LiDAR Optical Precision Using Varioptic Liquid Lenses and a Two-Lens System.*
- **Joint Mathematics Meeting (Jan 2024):** *Using the Gini Coefficient to Rank AI Models.*